Direct and indirect beams of interstellar neutral helium as a probe of the helium ionization rate

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Abstract

Interstellar neutral (ISN) helium reaches 1 au on two families of Keplerian orbits: a *direct* beam and an *indirect* beam that wraps around the Sun before arriving. Because indirect atoms spend longer close to the Sun, the ratio of the two beam fluxes depends on the ionization rate but not on the interstellar density or on the absolute detection efficiency. I implement an analytic hot model of ISN He at 1 au and use it to predict both beams over the year as seen by a Sun-pointing spinning detector such as IBEX-Lo or IMAP-Lo. For a reference ionization rate $\beta_0 = 1.0 \times 10^{-7}\,\mathrm{s}^{-1}$ at 1 au, a 43% higher rate (as suggested by New Horizons/SWAP pickup ions) lowers the ratio of the late-October indirect beam to the February direct beam by a factor of 0.61, while a 43% higher density leaves it unchanged. This report describes the model, the assumed observing conditions and the resulting figures, and lists the physics and instrument effects that still need to be included.

1 Introduction

The Sun moves through the partially ionized local interstellar medium, and its neutral component flows freely into the heliosphere. Helium is the best-studied ISN species: it has a high ionization potential, it is only weakly filtered at the heliospheric boundary, and it is too heavy for radiation pressure to matter much, so its trajectories are close to Keplerian hyperbolae. The flow was measured directly by the GAS instrument on Ulysses (Witte, 2004) and, since 2009, by IBEX-Lo on the Interstellar Boundary Explorer (McComas et al., 2009; Fuselier et al., 2009; Möbius et al., 2009). Analyses of the IBEX-Lo data (Bzowski et al., 2012; Lee et al., 2012) led to a consensus set of inflow parameters (speed, direction and temperature; McComas et al. 2015). IBEX also revealed a slower, warmer secondary population, the "Warm Breeze" (Kubiak et al., 2014).

The main loss process for ISN He inside the heliosphere is ionization, mostly by solar EUV photons with a contribution from electron impact (Sokół et al., 2019). The ionization rate controls how much helium survives to 1 au and therefore enters every interpretation of the data. There are now two independent hints that the rate is higher than solar-proxy models predict. Swaczyna et al. (2022) found that the evolution of the ISN He flux seen by IBEX-Lo over a solar cycle requires either a change in detection efficiency, a higher ionization rate, or an additional signal source. Antonik et al. (2025) derived the ionization rate from He^+ pickup ions measured by New Horizons/SWAP and found it to be ≈43% higher than the model rates.

The difficulty is that neither measurement can separate the ionization rate from the interstellar density on its own. Pickup-ion production scales as $n_{\mathrm{ISN}}\,\beta$, and the direct-beam flux at IBEX-Lo scales as $n_{\mathrm{ISN}}\,\varepsilon\,\exp(-\beta_0\langle X\rangle)$, where ε is the poorly known absolute efficiency and $\langle X\rangle$ is an exposure factor defined below. "More ionization", "more helium" and "a different efficiency" are therefore partly degenerate. Swaczyna et al. (2025) adopt an ISN He density of $\approx 0.01\,\mathrm{cm}^{-3}$ at 1 au during solar minimum, which is one of the comparison points used below.

Bzowski et al. (2023) proposed a calibration-free way to measure the ionization rate: compare the fluxes of the direct and indirect beams. Atoms on indirect orbits pass around the Sun before

reaching 1 au and accumulate more ionization exposure, so the flux ratio depends on β_0 while the density and the absolute efficiency cancel. An earlier attempt with Ulysses data, summarized by Bzowski et al. (2023), was limited because the indirect beam could not be separated from the background. The pivot platform of IMAP-Lo on the Interstellar Mapping and Acceleration Probe (McComas et al., 2018) makes the indirect beam observable, and Kubiak et al. (2023) designed a pivot-angle schedule that includes calibration-free determination of the He and Ne ionization rates.

This project is a lightweight, independent implementation of the direct/indirect-beam idea. It asks a narrow question: if the SWAP result is correct, how large is the change in the beam ratio, and does the ratio really distinguish a higher ionization rate from a higher density?

2 Methodology

2.1 Interstellar gas at infinity

Far from the Sun the ISN He is a drifting Maxwellian,

$$f_{\mathrm{LISM}}(\mathbf{v}) = \frac{n_{\mathrm{ISN}}}{\pi^{3/2} v_{\mathrm{th}}^3} \exp\left(-\frac{|\mathbf{v}-\mathbf{u}|^2}{v_{\mathrm{th}}^2}\right), \qquad v_{\mathrm{th}} = \sqrt{2k_B T/m_{\mathrm{He}}}, \tag{1}$$

with the IBEX consensus parameters of McComas et al. (2015): inflow speed $u = 25.4\,\mathrm{km\,s^{-1}}$, upwind ecliptic longitude $\lambda = 255.7°$, latitude $\beta = 5.1°$ and temperature $T = 7500\,\mathrm{K}$ ($v_{\mathrm{th}} \approx 5.6\,\mathrm{km\,s^{-1}}$). The density at infinity is $n_{\mathrm{ISN}} = 0.015\,\mathrm{cm^{-3}}$.

2.2 Keplerian orbits and the two beams

Radiation pressure is neglected, so each atom moves on a hyperbola under solar gravity alone (see Section 5 for the consequences). For a phase-space point $(\mathbf{r}, \mathbf{v})$ the orbit is fixed by the specific energy $E = v^2/2 - GM_\odot/r > 0$ and the specific angular momentum $h = |\mathbf{r} \times \mathbf{v}|$. The eccentricity is $e = \sqrt{1 + 2Eh^2/(GM_\odot)^2}$ and the incoming asymptote lies at true anomaly $-\nu_\infty$ with $\nu_\infty = \arccos(-1/e)$. The angle the atom has swept since infinity is $\Delta\nu = \nu + \nu_\infty$, where ν is its current true anomaly. Two hyperbolae with the same speed at infinity pass through every point near the Sun:

- the *direct* orbit, with $\Delta\nu < 180°$;
- the *indirect* orbit, which passes around the Sun first and has $\Delta\nu > 180°$.

The speed at infinity is $v_\infty = \sqrt{2E}$, and its direction follows from rotating the position along the orbit back to the asymptote. All of this is done analytically, without integrating the equations of motion.

2.3 Ionization exposure

The ionization rate is assumed to fall off as $\beta(r) = \beta_0\,(r_0/r)^2$ with $r_0 = 1\,\mathrm{au}$. Along a Keplerian orbit the true anomaly changes as $\mathrm{d}\nu/\mathrm{d}t = h/r^2$, so the integrated loss has a closed form:

$$\int_{-\infty}^{t} \beta\,\mathrm{d}t' = \beta_0 r_0^2 \int \frac{\mathrm{d}t'}{r^2} = \beta_0\,\frac{r_0^2\,\Delta\nu}{h} \equiv \beta_0 X. \tag{2}$$

The survival probability of an atom is $\exp(-\beta_0 X)$. Indirect atoms have both a larger $\Delta\nu$ and, typically, a smaller h, so their exposure X is several times larger than that of direct atoms reaching the same point. Equation (2) is exact for a pure r^{-2} rate on a Keplerian orbit, and the test suite checks it against numerical orbit integration.

2.4 Distribution function and moments

By Liouville's theorem the distribution function is conserved along orbits, apart from the ionization losses, so at any point inside the heliosphere

$$f(\mathbf{r},\mathbf{v}) = f_{\mathrm{LISM}}(\mathbf{v}_\infty(\mathbf{r},\mathbf{v}))\,\exp(-\beta_0 X(\mathbf{r},\mathbf{v})). \tag{3}$$

This is the classical "hot model" of interstellar gas in the heliosphere (Thomas, 1978; Lee et al., 2012). The density is $n = \int f\,\mathrm{d}^3v$. For an observer moving with velocity $\mathbf{v}_{\mathrm{obs}}$, an atom arriving from direction $\hat{\mathbf{d}}$ with speed s relative to the observer has heliocentric velocity $\mathbf{v} = \mathbf{v}_{\mathrm{obs}} - s\hat{\mathbf{d}}$, and the sky-integrated number flux is

$$F = \iint f(\mathbf{r}, \mathbf{v}_{\mathrm{obs}} - s\hat{\mathbf{d}})\, s^3\,\mathrm{d}s\,\mathrm{d}\Omega. \tag{4}$$

Restricting the integral to $\Delta\nu < 180°$ or $\Delta\nu > 180°$ gives F_{dir} and F_{ind}.

The integrals are evaluated with the midpoint rule on a grid of arrival directions (2° pixels) times speeds (1 km s^{-1} steps up to 120 km s^{-1}); the sky maps use a 1° and 0.5 km s^{-1} grid. Cells with $|\mathbf{v}_\infty - \mathbf{u}|^2/v_{\mathrm{th}}^2 > 30$ are dropped. Each remaining cell is stored once together with its X, $\Delta\nu$ and energy, so any β_0, density or energy threshold can be applied afterwards without recomputing the orbits. Halving and quartering the grid step changes the beam fluxes by less than 10^{-4} in most seasons and by about 10^{-3} where the beams overlap.

2.5 Why the ratio measures the ionization rate

Each beam is a sum of exponentials in β_0, so its flux-weighted mean exposure is

$$\langle X\rangle = -\frac{\mathrm{d}\ln F}{\mathrm{d}\beta_0}. \tag{5}$$

To first order in β_0,

$$\ln\frac{F_{\mathrm{ind}}}{F_{\mathrm{dir}}} = \mathrm{const} - \beta_0(\langle X\rangle_{\mathrm{ind}} - \langle X\rangle_{\mathrm{dir}}). \tag{6}$$

Both fluxes are proportional to n_{ISN} and, for the same instrument, to the efficiency ε, so both cancel in the ratio. A change of β_0 moves the ratio, while a change of n_{ISN} does not. The cancellation of ε is exact only if the efficiency is the same for both beams; for beams arriving at different energies an energy-dependent efficiency leaves a residual (Section 5).

3 Instrumentation

Observation. The observer is on a circular orbit at 1 au in the ecliptic plane, moving prograde at 29.78 km s^{-1}. Its position is given by its ecliptic longitude; the conversion to calendar date uses a low-precision solar ephemeris (Earth at 30° in late October, 130° in early February). The annual tables are computed every 2.5° of longitude, and on a grid of β_0 from 0 to 3×10^{-7} s^{-1} in steps of 5×10^{-9} s^{-1}, with interpolation in $\ln F$ between grid points.

Detector geometry. IBEX is a Sun-pointing spinning spacecraft, and IBEX-Lo looks perpendicular to the spin axis, so in each orbit it samples a great circle 90° from the Sun (McComas et al., 2009; Fuselier et al., 2009). This geometry sees the direct He beam well around February, which is where most IBEX ISN He analyses come from (Möbius et al., 2009; Bzowski et al., 2012). IMAP-Lo is mounted on a pivot platform that changes the angle between the boresight and the spin axis (McComas et al., 2018), so it can scan circles of other radii around the Sun, including ones that cross the indirect beam (Kubiak et al., 2023). In the model a "scan circle" is the set of directions at a given angle from the Sun direction, and the angle needed to see a beam is the angle between its flux-weighted centroid and the Sun.

Detection. IBEX-Lo and IMAP-Lo detect He indirectly, through negative ions sputtered from a conversion surface, so the efficiency depends on energy and falls off at low energy. The model mostly uses sky-integrated fluxes without an instrument response. The energy dependence is represented only by a sharp lower threshold of 0, 20 or 40 eV in the observer frame.

Reference observations. The direct beam is taken at its February peak (Earth at 130°), where IBEX-Lo has measured it for over a decade. The indirect beam is taken in late October (Earth at 30°), where it is bright, well separated from the direct beam and about 59° from the Sun. The two competing explanations of the SWAP result are

- **H1**: the ionization rate is 1.43× the model value, $\beta_0 = 1.43 \times 10^{-7}\,\mathrm{s}^{-1}$, at fixed density;
- **H2**: the density is 1.43× higher at the model rate $\beta_0 = 1.0 \times 10^{-7}\,\mathrm{s}^{-1}$, giving the same pickup-ion production $n_{\mathrm{ISN}}\beta$.

4 Results

Table 1 collects the main numbers. The figures are described below in order.

Table 1: Summary for $\beta_0 = 1.0 \times 10^{-7}\,\mathrm{s}^{-1}$, direct beam at Earth longitude 130° (February) and indirect beam at 30° (late October).

	Direct (Feb.)	Indirect (late Oct.)
Sky-integrated flux ($\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$)	9.2×10^4	1.1×10^4
Mean energy, observer frame (eV)	129	106
$\langle X \rangle$ (10^7 s)	0.67	1.83
Beam centre (ecliptic lon, lat)	(220°, 3.9°)	(151°, −3.1°)
RMS beam width	8.1°	7.2°
Angle from the Sun	90°	59°
Flux change for H1 ($\beta_0 \times 1.43$)	0.75	0.46
Flux change for H2 ($n_{\mathrm{ISN}} \times 1.43$)	1.43	1.43

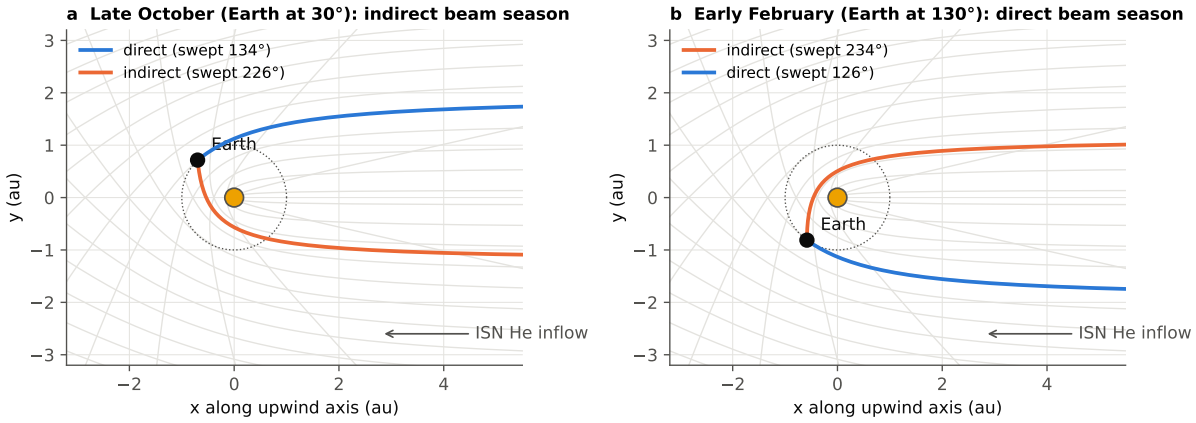


Figure 1: Cold-model ($T = 0$) sketch of the direct (blue) and indirect (orange) orbits reaching Earth in late October (a) and early February (b), in the plane of the inflow. Grey lines are other orbits of the cold flow, the dotted circle is Earth's orbit, and the gas arrives from the right. The swept angle $\Delta\nu$ of each orbit is given in the legend.

Trajectories (Figure 1). For a cold gas the impact parameters of the two orbits through a point at distance r and polar angle ψ from the upwind axis solve $b^2 - r\sin\psi\, b - (GM_\odot r/u^2)(1-\cos\psi) = 0$. In late October the direct orbit reaching Earth has swept 134° and the indirect one 226°; in February the values are 126° and 234°. The indirect orbits pass much closer to the Sun, which is why they accumulate more exposure.

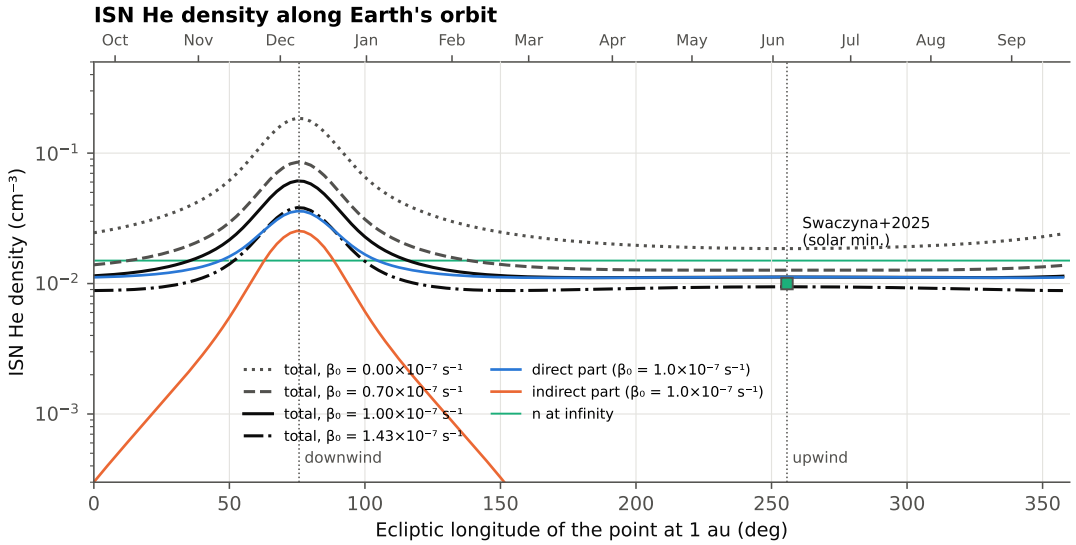


Figure 2: ISN He density along Earth's orbit for several ionization rates (black and grey), and the direct and indirect parts for $\beta_0 = 1.0 \times 10^{-7}\,\mathrm{s}^{-1}$ (blue, orange). The green line is the density at infinity, and the square is the solar-minimum value of Swaczyna et al. (2025).

Density at 1 au (Figure 2). This is a consistency check of the model. Upwind at 1 au the density is $0.0112\,\mathrm{cm}^{-3}$ for $\beta_0 = 1.0 \times 10^{-7}\,\mathrm{s}^{-1}$, close to the $\approx 0.01\,\mathrm{cm}^{-3}$ of Swaczyna et al. (2025). Without ionization it is $0.0185\,\mathrm{cm}^{-3}$, above the value at infinity because of gravitational focusing. Downwind (Earth longitude $\approx 75°$, early December) the focusing cone raises the density to $0.061\,\mathrm{cm}^{-3}$, about 40% of which is indirect atoms. Upwind, the indirect contribution is negligible ($< 10^{-7}$ of the total).

Sky maps (Figure 3). In late October the indirect beam is a compact spot near ecliptic longitude 151°, about 59° from the Sun. It is not crossed by the 90° IBEX-Lo circle, but a 59° pivot angle passes through its centre. The direct population in October is a broad, slow component that overtakes Earth from behind. In February the direct beam is compact and sits on the IBEX-Lo circle at longitude $\approx 220°$, while the indirect beam comes from within a few degrees of the Sun and cannot be observed.

The beams over the year (Figure 4). The direct beam is bright all year, peaking at $\approx 2.1 \times 10^5\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}$ when Earth passes through the focusing cone. The indirect beam is comparable to the direct beam from November to December. It becomes negligible around the upwind season (May to July), when Earth is far from the region where indirect orbits arrive. Between Earth longitudes of about 60° and 90° the beams overlap on the sky and cannot be separated. The indirect beam is more sensitive to β_0 throughout. Its energy in autumn is 100 eV to 110 eV, compared with ≈ 129 eV for the February direct beam. The two reference observations therefore differ by about 20% in energy, so an energy-dependent efficiency cancels only partly in their ratio. Bzowski et al. (2023) recommend choosing pairs of observations at matched energies

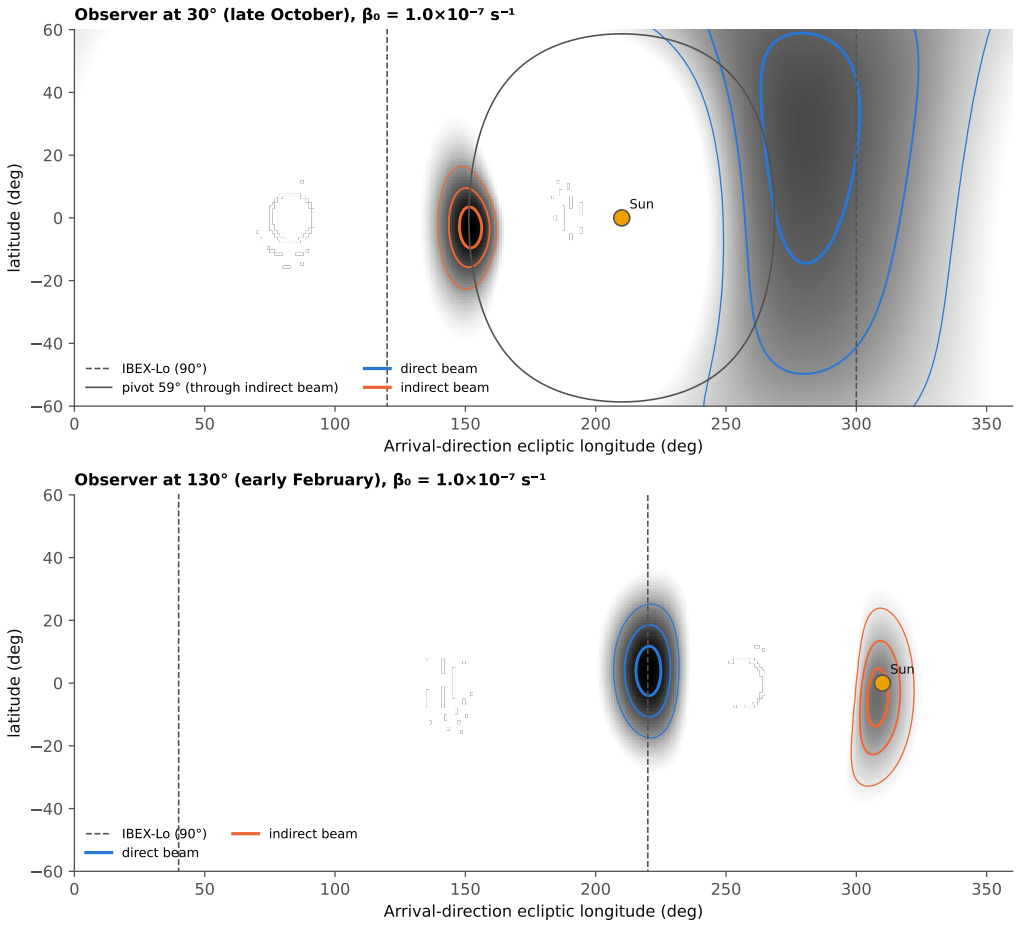


Figure 3: Directional flux in the observer frame (arrival directions, ecliptic coordinates) for Earth at 30° (top) and 130° (bottom). Grey shading is the total flux; blue and orange contours mark 1, 10 and 50% of the peak of the direct and indirect beams. The dashed curve is the IBEX-Lo scan circle (90° from the Sun) and the solid curve in the top panel is the pivot angle that crosses the indirect-beam centre.

for this reason. Panel (c) shows that the indirect beam stays 50°–60° from the Sun in autumn, so it needs a pivot angle smaller than 90°.

Exposure and response (Figure 5). The indirect beam accumulates 1.5–5 times more exposure than the direct beam observed at the same time, with the largest contrast early in autumn: the ratio $\langle X\rangle_{\rm ind}/\langle X\rangle_{\rm dir}$ is 4.7 at Earth longitude 0°, 2.6 at 30° and 1.5 at 55°. For the reference pair, $\langle X\rangle$ is 0.67×10^7 s for the February direct beam and 1.83×10^7 s for the October indirect beam. Under H1 the direct beam drops to 0.75 and the indirect beam to 0.46 of the model value, so the ratio drops to 0.61. Under H2 both beams rise by the same factor of 1.43 and the ratio does not change. The absolute direct-beam flux could separate H1 and H2 only if the absolute efficiency were known to much better than the 25–43% difference between them; the ratio does not need it.

Breaking the degeneracy (Figure 6). The pseudo-data assume 10^4 counts in the February direct beam, and the October indirect-beam counts (≈ 750) follow from the same exposure time and efficiency. A single direct-beam measurement constrains only a curved band in the $(\beta_0, n_{\rm ISN}\varepsilon)$ plane: a higher ionization rate can be compensated by more helium or a higher efficiency. Adding the indirect beam closes the contour around the true point. H2, with its

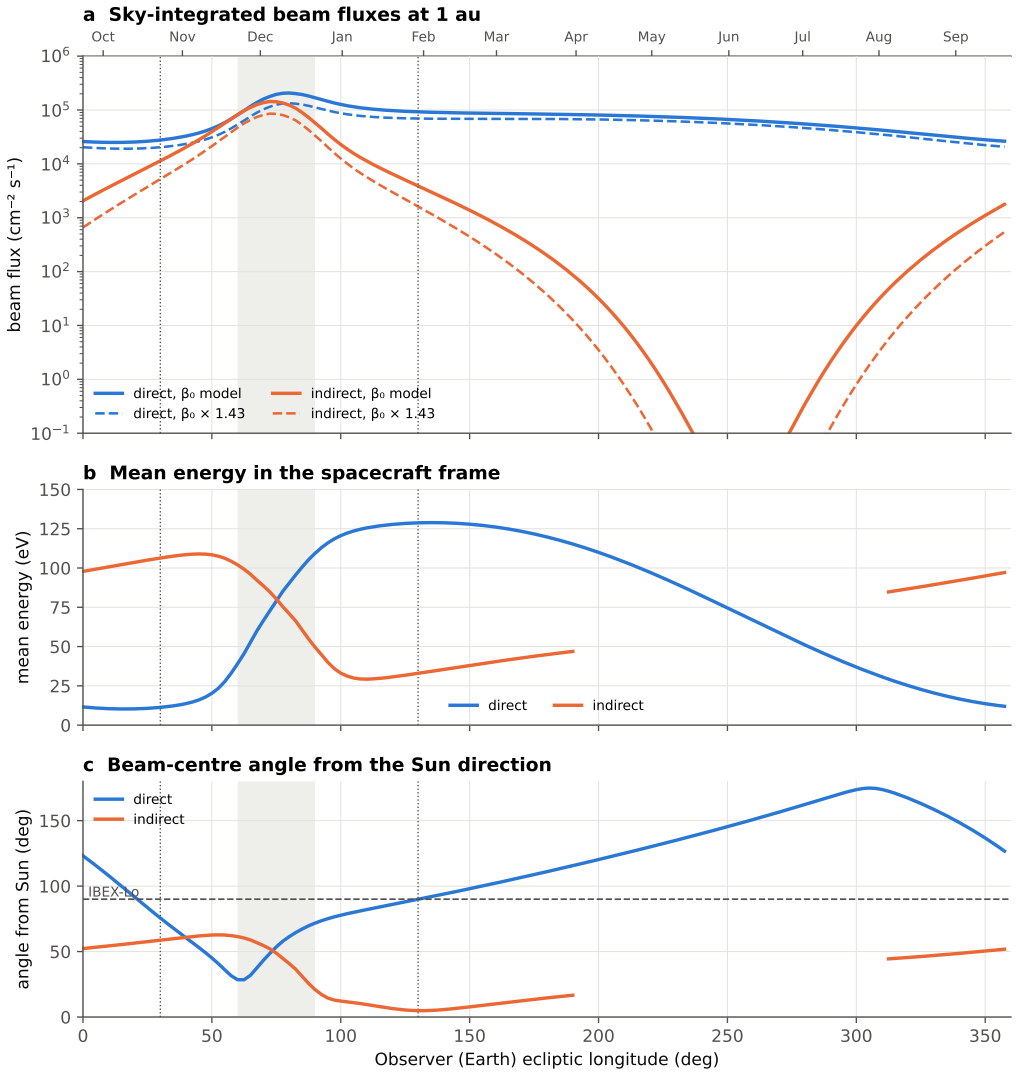


Figure 4: The two beams over the year versus Earth's ecliptic longitude (calendar months on top). (a) Sky-integrated fluxes for the model rate (solid) and $\beta_0 \times 1.43$ (dashed). (b) Flux-weighted mean energy in the observer frame. (c) Angle between the beam centre and the Sun. Grey bands mark where the beam centres are closer than the sum of their RMS widths. The dotted lines mark the reference seasons (30° and 130°). The indirect beam is not drawn in (b) and (c) where its flux is below 0.1% of the direct beam's.

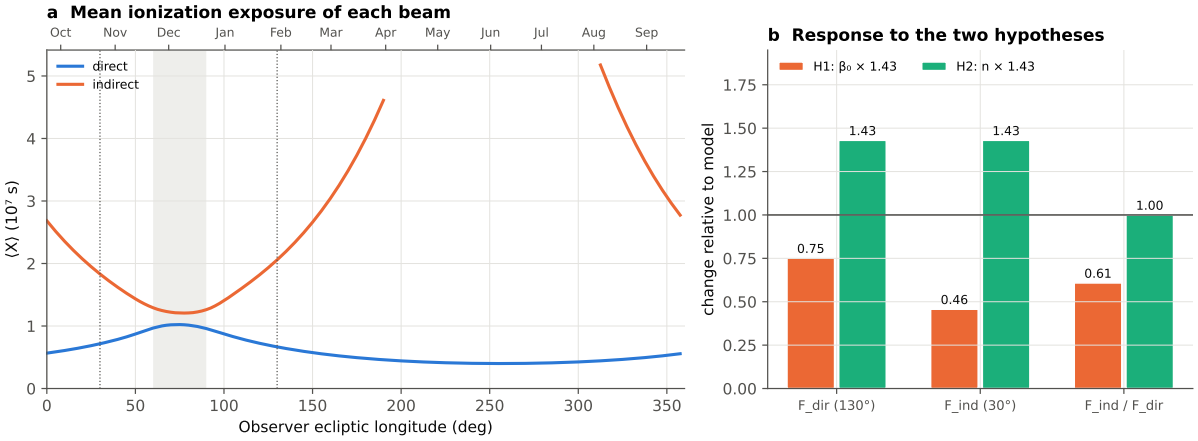


Figure 5: (a) Flux-weighted mean exposure $\langle X \rangle$ of each beam over the year; grey band as in Figure 4. (b) Response of the February direct-beam flux, the late-October indirect-beam flux and their ratio to H1 ($\beta_0 \times 1.43$, orange) and H2 ($n_{\mathrm{ISN}} \times 1.43$, green), relative to the model.

best-fitting $n_{\mathrm{ISN}}\varepsilon$, lies at $\Delta\chi^2 \approx 199$ from the truth. This is an idealized illustration of how the second beam lifts the degeneracy, not a forecast of the achievable precision, which depends on the effects in Section 5.

Energy threshold (Figure 7). At Earth longitude 30° the indirect beam (mean energy $\approx 106\,\mathrm{eV}$) is unaffected by a threshold of up to 40 eV. The slow direct population seen in autumn ($\approx 10\,\mathrm{eV}$ to $20\,\mathrm{eV}$) drops from 2.8×10^4 to $8.3 \times 10^2\,\mathrm{cm^{-2}\,s^{-1}}$ for $E > 20\,\mathrm{eV}$, and to $\approx 7\,\mathrm{cm^{-2}\,s^{-1}}$ for $E > 40\,\mathrm{eV}$. A realistic low-energy efficiency cut-off therefore helps by suppressing the slow direct atoms that share the sky with the indirect beam in autumn. The February direct beam is unaffected by these thresholds.

5 Limitations and future prospects

The model is a feasibility study of the size of the beam-ratio signal. A full analysis of IMAP-Lo data would need several additions, listed roughly in order of importance. Some of them are already being worked on.

1. **Radiation pressure.** The orbits are purely Keplerian. Kowalska-Leszczyńska et al. (2023) showed that radiation pressure on He changes the IMAP-Lo fluxes by up to ∼9% for the indirect beam, likely more than the statistical uncertainties. Including it breaks the closed form of Equation (2); the natural upgrade is numerical orbits with a radiation-pressure factor.

2. **Other near-Sun losses.** The beam ratio measures the *total* extra loss of indirect atoms, not photoionization alone. Any process that removes atoms preferentially close to the Sun lowers the ratio in the same way as a higher β_0. Elastic scattering of ISN He by solar-wind particles has been proposed as an alternative to a ∼40% higher ionization rate (arXiv:2409.11784, 2024). Separating the two would rely on their different energy and angular signatures.

3. **Secondary helium.** The Warm Breeze (Kubiak et al., 2014) arrives from a different direction with its own direct and indirect beams, and its contribution to the autumn indirect beam has to be modelled. The code supports this by computing a second population with different inflow parameters and adding the fluxes.

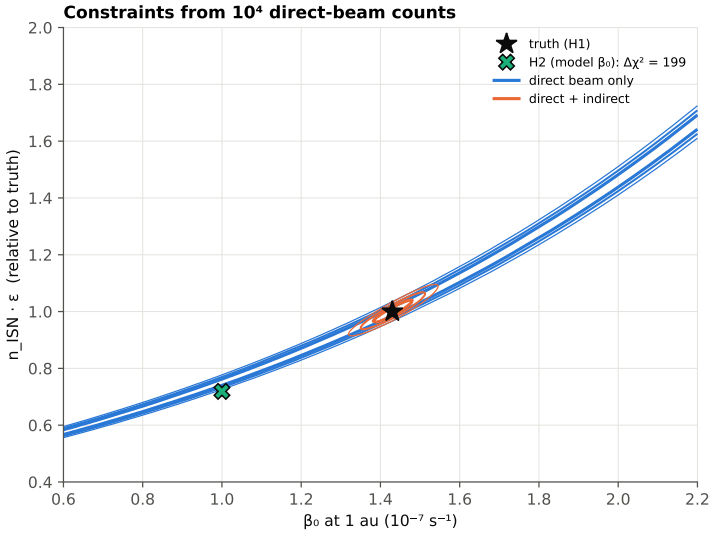


Figure 6: Constraints in the plane of β_0 and the product $n_{\mathrm{ISN}}\,\varepsilon$ (relative to the truth) from noise-free pseudo-data generated under H1 (star). Blue: the February direct beam alone; orange: direct plus late-October indirect beam. Contours are $\Delta\chi^2 = 2.30$, 6.18 and 11.8 of the Poisson likelihood of Baker & Cousins (1984). The cross marks H2 with $n_{\mathrm{ISN}}\varepsilon$ fitted to both beams.

4. **Time-dependent and non-r^{-2} ionization.** Indirect atoms spend months near the Sun, and the two beams are observed months apart, so solar-cycle variations of the ionization rate matter. Electron-impact ionization falls off faster than r^{-2} and is not described by a single β_0 (Sokół et al., 2019). Both effects break the closed form; a numerical exposure integral along each orbit is the planned upgrade.

5. **Instrument response and energy matching.** The model uses sky-integrated fluxes and a sharp energy threshold. It does not include the IMAP-Lo collimator, the pivot schedule of Kubiak et al. (2023), spin-axis offsets, the energy-dependent sputtering efficiency or the duty cycle. Choosing direct/indirect pairs at matched energies (Bzowski et al., 2023) would reduce the efficiency residual.

6. **Inflow-parameter systematics and counting statistics.** The uncertainties of the inflow speed, direction and temperature (McComas et al., 2015) also move the beam ratio, and the number of counts needed to detect the signal depends on the observing season. A first estimate was removed from this version and is being redone. Because the inflow parameters are correlated and are themselves derived from direct-beam data, the consistent approach is a joint fit of the inflow parameters and β_0 to both beams.

7. **Geometry.** IMAP's position near L1 (0.99 au), the eccentricity of Earth's orbit and the filtration of He in the outer heliosheath are ignored. These are expected to be small corrections.

Code and data

The model, the scripts that generate the data products and the notebook that makes the figures are in the project repository (`src/isnhe`, `scripts/` and `notebooks/plot_figures.ipynb`).

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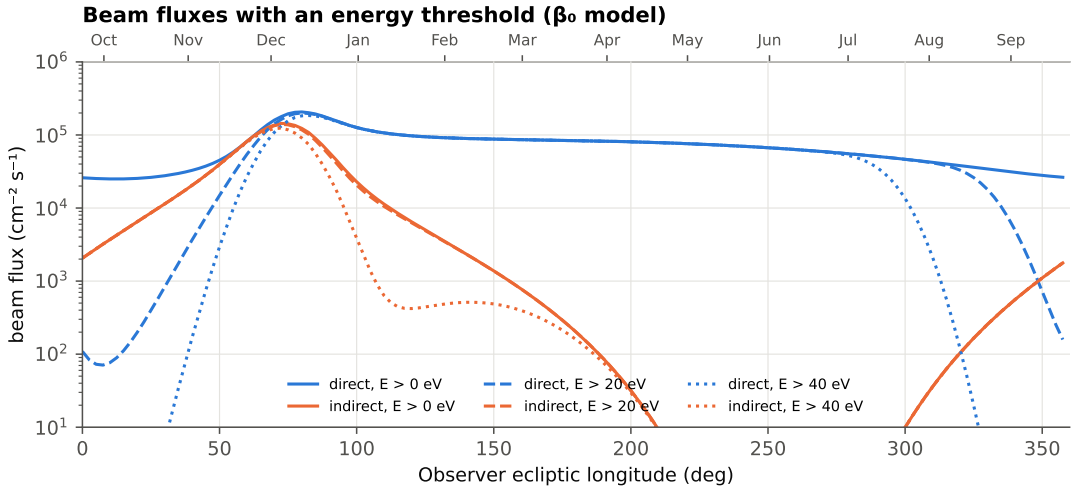


Figure 7: Beam fluxes for $\beta_0 = 1.0 \times 10^{-7}\,\mathrm{s}^{-1}$ with a sharp energy threshold in the observer frame of 0 (solid), 20 (dashed) and 40 eV (dotted).

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